

Assignment Report

Title: **Compiler Principles**

The Seventh Theory Homework

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[ABSTRACT]

After learning the lecture 7 of compiler principles[1], we got the homework at the end of the lecture.

There are some requirements for the homework:

- The file should be named as **“学号 _ 姓名 _ 第 i 次理论作业”**, where i is the number of the homework.
- The file should be in docx or **pdf** format.
- All the homework should be sent to the TA's email: **406648896@qq.com**
- The deadline for the homework is **24:00 on Sunday of the week** when the homework is assigned.

Keywords: Compiler Principles

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1 Exercise 7

1.1 Exercise 7.1

1.1.1 Problem

Consider the grammar

$$S \rightarrow (\ S\ R\ \mid\ a,$$

$$R \rightarrow,\ S\ R\ \mid\ )$$

Try to construct an SLR(1) parsing table for the grammar, and check whether there are conflicts in the parsing table.

1.1.2 Solution

No conflicts are found in the SLR(1) parsing table, so the grammar is SLR(1).

$$\begin{array}{ll} 0 & S' \rightarrow S \\ 1 & S \rightarrow (\ S\ R \\ 2 & S \rightarrow a \\ 3 & R \rightarrow,\ S\ R \\ 4 & R \rightarrow) \end{array}$$

$$\begin{aligned} \text{FIRST}(S) &= \{(,\ a\} \\ \text{FIRST}(R) &= \{,,\ )\} \\ \text{FOLLOW}(S) &= \{\$,\ ,,\ )\} \\ \text{FOLLOW}(R) &= \{\$,\ ,,\ )\} \end{aligned}$$

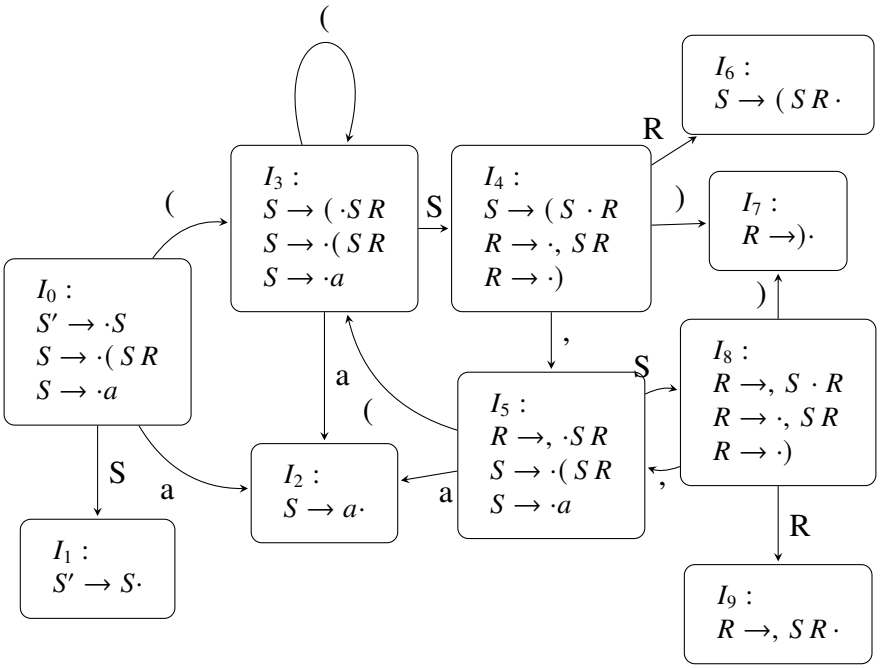


Figure 1.1 Canonical LR(0) DFA (states I_0–I_9) for Exercise 7.1.

State	ACTION					GOTO	
	(	,	a	)	$	S	R
0	S3		S2			1	
1					acc		
2		r2		r2	r2		
3	S3		S2			4	
4		S5		S7			6
5	S3		S2			8	
6		r1		r1	r1		
7		r4		r4	r4		
8		S5		S7			
9		r3		r3	r3		

Table 1.1 SLR(1) parsing table for Exercise 7.1

1.2 Exercise 7.2

1.2.1 Problem

Consider the grammar

$$S \to S\,a\,b \mid a\,R,$$

$$R \to S \mid a.$$

Is this grammar SLR(1)? Explain why or why not.

1.2.2 Solution

$$\begin{array}{ll} 0 & S' \to S \\ 1 & S \to S\,a\,b \\ 2 & S \to a\,R \\ 3 & R \to S \\ 4 & R \to a \end{array}$$

$$\begin{aligned} \text{FIRST}(S) &= \{a\} \\ \text{FIRST}(R) &= \{a\} \\ \text{FOLLOW}(S) &= \{a,\ \$\} \\ \text{FOLLOW}(R) &= \{a,\ \$\} \end{aligned}$$

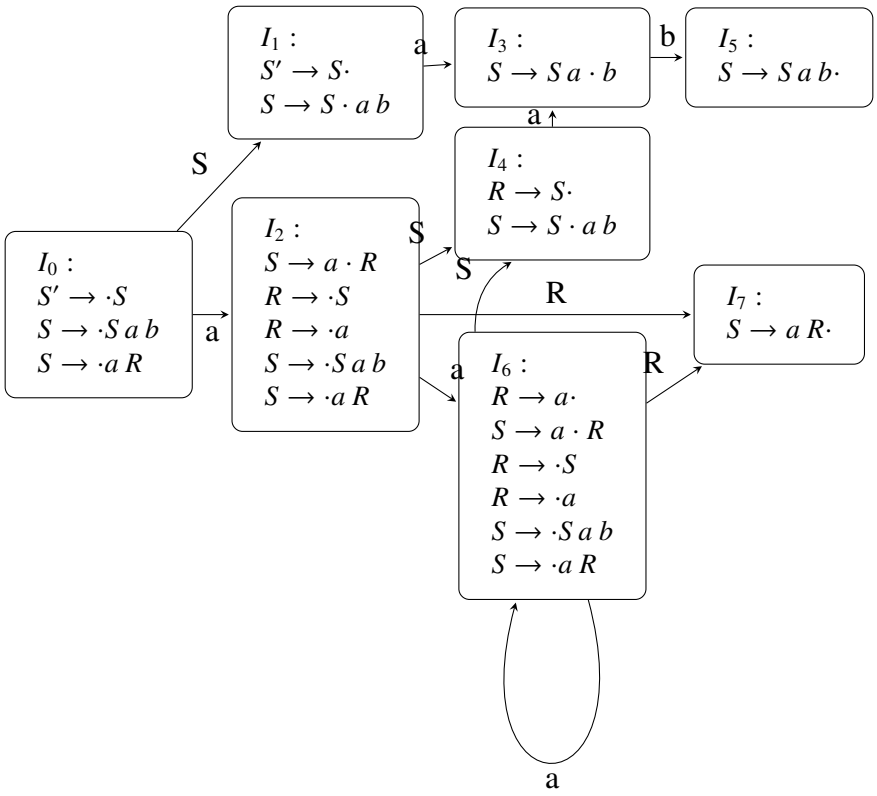


Figure 1.2 Canonical LR(0) DFA (states I_0–I_7) for Exercise 7.2.

结论 In state I_4 (and also in I_6), there exist both shift and reduce actions for input a (for example, `S3` and `r3` conflict in I_4), therefore the grammar is not SLR(1).

1.3 Exercise 7.3

1.3.1 Problem

Consider the grammar

$$S \to A,$$

$$A \to B\,A \mid \varepsilon,$$

$$B \to a\,B \mid b.$$

(1) Prove that the grammar is an LR(1) grammar.

(2) Construct an LR(1) parsing table for the grammar.

(3) Show the detailed parsing procedure for the sentence **abab**, following the style in the lecture slides.

1.3.2 Solution

Conclusion: By construction of the LR(1) automaton and parsing table, and verification via parsing trace, the grammar is **LR(1)**

The grammar is LR(1), as confirmed by the absence of conflicts in the LR(1) parsing table.

$$\begin{aligned}
&0 \quad S' \to S\\
&1 \quad S \to A\\
&2 \quad A \to B\,A\\
&3 \quad A \to \varepsilon\\
&4 \quad B \to a\,B\\
&5 \quad B \to b
\end{aligned}$$

$$\begin{aligned}
\text{FIRST}(S) &= \{\texttt{a},\ \texttt{b},\ \varepsilon\}\\
\text{FIRST}(A) &= \{\texttt{a},\ \texttt{b},\ \varepsilon\}\\
\text{FIRST}(B) &= \{\texttt{a},\ \texttt{b}\}
\end{aligned}$$

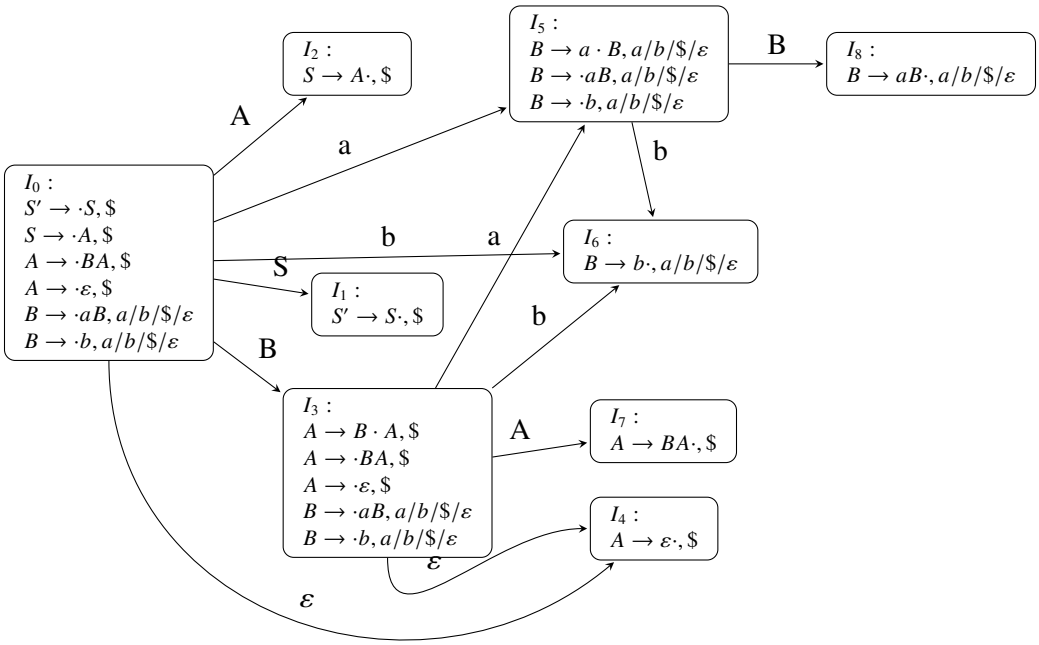


Figure 1.3 Canonical LR(1) DFA for the extended grammar (states I_0–I_8).

State	ACTION				GOTO		
	a	b	ε	$	S	A	B
0	S5	S6	S7		1	2	3
1				acc			
2				r1			
3	S5	S6	S4			7	3
4				r3			
5	S5	S6					8
6	r5	r5	r5	r5			
7				r2			
8	r4	r4	r4	r4			

Table 1.2 LR(1) parsing table

Step	Symbol	State	Input	Reference	Action	Output
1	\$	0	abab\$	$a[0,a] = S5$	shift	
2	\$ a	05	bab\$	$a[5,b] = S6$	shift	
3	\$ ab	056	ab\$	$a[6,\varepsilon] = r5\ g[5,B] = 8$	reduce	$B \to b$
4	\$ aB	058	ab\$	$a[8,a] = r4\ g[0,B] = 3$	reduce	$B \to aB$
5	\$ aB a	03	ab\$	$a[3,a] = S5$	shift	
6	\$ aB ab	035	b\$	$a[5,b] = S6$	shift	
7	\$ aB aB	0356	\$	$a[6,\$] = r5\ g[5,B] = 8$	reduce	$B \to b$
8	\$ aB AB	0358	\$	$a[8,\$] = r4\ g[3,B] = 3$	reduce	$B \to aB$
9	\$ aB AB	033	ε \$	$a[3,\varepsilon] = S4$	shift	
10	\$ aB A	0334	\$	$a[4,\varepsilon] = 13\ g[3,A] = 7$	reduce	$A \to \varepsilon$
11	\$ aB A	0337	\$	$a[7,\$] = r2\ g[3,A] = 7$	reduce	$A \to BA$
12	\$ aB	037	\$	$a[7,\$] = r2\ g[0,A] = 2$	reduce	$A \to BA$
13	\$ aB	02	\$	$a[2,\$] = r1\ g[0,S] = 1$	reduce	$S \to A$
14	\$ S	01	\$	$a[1,\$] = acc$	accept	

Table 1.3 Parsing steps for input `abab$`

1.4 Exercise 7.4*

1.4.1 Problem

Consider the grammar (from DBv2, ex.4.7.4):

$$S \to A\,a \mid b\,A\,c \mid d\,c \mid b\,d\,a,$$
$$A \to d.$$

Show that the grammar is LALR(1) but not SLR(1).

1.4.2 Solution

Conclusion: In the above LR(1) transition diagram, there are no compatible (mergeable) core-equivalent item sets, and no shift–reduce or reduce–reduce conflicts exist; therefore, the grammar is LALR(1). For state I_4, a shift–reduce conflict exists: $\text{FOLLOW}(A) = \{a, c\}$ and $A \to d$, so $\text{ACTION}[4, a] = \text{r5}$, while $S \to d \cdot c$ implies $\text{ACTION}[4, a] = \text{S8}$. Hence, the grammar is **not** SLR(1).

It is LALR(1), but not SLR(1).

The grammar is LR(1), and the canonical LR(1) automaton consists of 11 states (I_0 to I_{11}), as shown below.

$$
\begin{aligned}
0 \quad & S' \to S \\
1 \quad & S \to A\,a \\
2 \quad & S \to b\,A\,C \\
3 \quad & S \to d\,c \\
4 \quad & S \to b\,d\,a \\
5 \quad & A \to d
\end{aligned}
$$

$$
\begin{aligned}
\text{FIRST}(S) &= \{\texttt{b},\ \texttt{d}\} \\
\text{FIRST}(A) &= \{\texttt{d}\} \\
\text{FOLLOW}(S) &= \{\$\} \\
\text{FOLLOW}(A) &= \{\texttt{a},\ \texttt{c}\}
\end{aligned}
$$

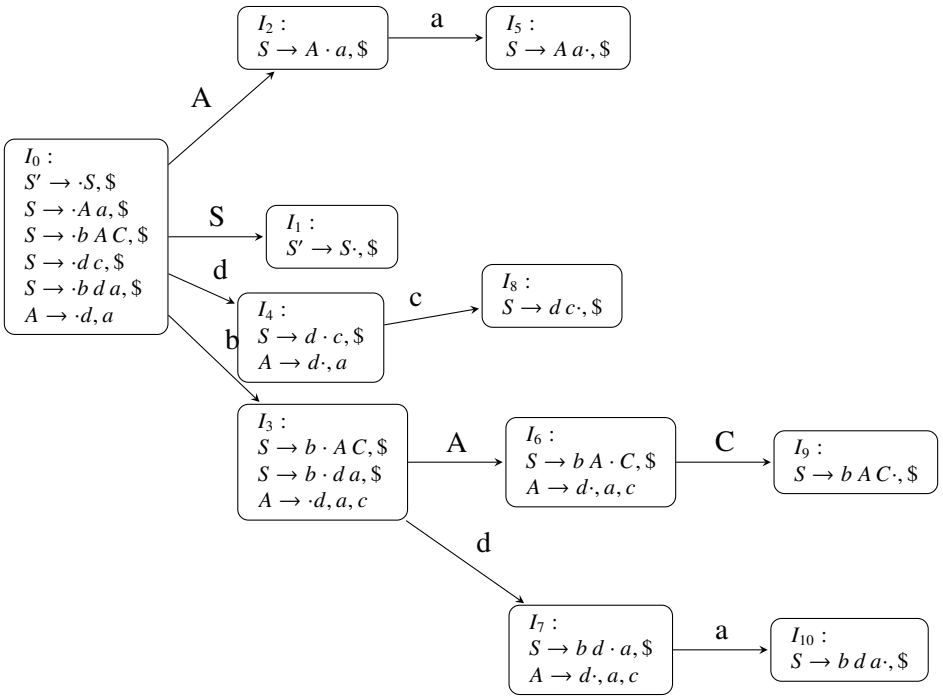


Figure 1.4 Canonical LR(1) DFA (states I_0–I_{10}) for Exercise 7.4.

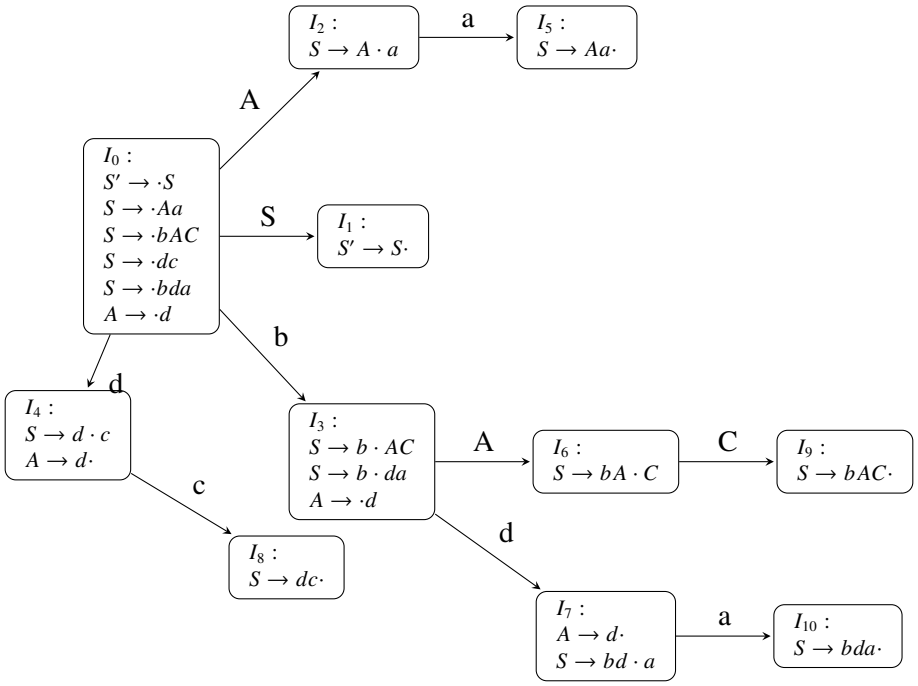


Figure 1.5 SLR(1) DFA (core items only). State I_4 has a shift-reduce conflict on symbol c (since c $\in$ FOLLOW(A) and $S \to d \cdot c$ requires shift on c).

References

[1] JUN LI W. Lecture07. principles of compiler construction[Z]. 2026.